

Question	Answer	Marks
5(a)	arrow from sphere, perpendicular to string, pointing left and down	B1
5(b)(i)	amplitude = 0.016 m	A1
5(b)(ii)	angular frequency = $2\pi / T$	C1
	$= 2\pi / 0.40$ $= 16 \text{ rad s}^{-1}$	A1
5(b)(iii)	total energy = $\frac{1}{2}m\omega^2x_0^2$	C1
	$= \frac{1}{2} \times 0.15 \times 15.7^2 \times 0.016^2$ $= 4.7 \times 10^{-3} \text{ J}$	A1
5(c)	dome-shaped curve starting and ending on the x-axis, with peak at $x = 0$	B1
	maximum E_k shown as $4.7 \times 10^{-3} \text{ J}$	B1
	minimum x shown as -0.016 m and maximum x shown as $+0.016 \text{ m}$ at the ends of the line	B1

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